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Sentinel-1 time series data for monitoring the phenology of winter wheat

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ABSTRACT

The potential of time series metrics based on optical sensors for monitoring of crops and their phenological stages has long been recognized. SAR (synthetic aperture radar) sensors such as Sentinel-1 can provide dense time series, but their potential for retrieving time series metrics is to date underexplored. The backscatter coefficient in VH (vertical-horizontal) and VV (vertical-vertical) polarization as well as the interferometric coherence of Sentinel-1 was retrieved in a German agricultural area in order to obtain a time series with a temporal resolution of 6 days. The backscatter coefficient ratio of VH and VV polarization was also calculated. Local extrema and breakpoints were detected in the locally smoothed time series data in wheat fields for the years 2017, 2018 and 2019. Breakpoints in the VH/VV ratio time series have high potential for detecting the date of shooting and harvesting of the studied wheat fields. The root mean squared differences between detected date and in situ observed date were 4.6 days in 2017, 5.3 days in 2018 and 9.5 days in 2019 for the shooting and 5.1 days in 2017, 8.2 days in 2018 and 10.4 days in 2019 for the harvest. Ripeness stages could not be detected with a high degree of accuracy. The results suggest that the VH/VV ratio has a high potential for monitoring the phenological stages of wheat fields, which can be used for large-scale crop productivity monitoring and risk assessment in agricultural cropping systems.

1. Introduction

The demand for food is increasing along with the worldwide increase in population (FAO, 2017). Agricultural cultivation must fulfil the demand for food production. At the same time, agricultural production should maintain biodiversity, material cycles and other ecosystem services and limit its environmental impact (Foley et al., 2005). Furthermore, an adaption of agricultural practices to potentially changing environmental conditions is a prerequisite for sustaining the food production together with other ecosystem services (Trnka et al., 2014; Veloso et al., 2017). Information about the phenological status, especially the shooting, but also ripeness and harvest, is of high significance for the monitoring of the ecosystem services (Conradt et al., 2015; Möller et al., 2017; Möller et al., 2019; Huang et al., 2019). This information about the dates of different phenological stages can be used in crop yield prediction and thus, it is important for insurance as well as adaption strategies (Conradt et al., 2015; Möller et al., 2017, 2019).

Remote sensing could be a reliable tool for detecting the phenological status of crops on a large scale. In addition, the combination of vegetation indices and phenological stages detected by remote sensing together with meteorological information improved the assessment of weather conditions and their impact on the plants and crop yields (Möller et al., 2019; Huang et al., 2019). It was also demonstrated that information about phenological status increased the accuracy of crop type maps (Bargiel, 2017; Li et al., 2019).

Optical remote sensing and its derived vegetation indices are frequently used to monitor agricultural areas (Arvor et al., 2011; Pan et al., 2015; Guo et al., 2016; Veloso et al., 2017). A high temporal resolution is important for monitoring the phenology of plants, especially when an accurate detection of the beginning of a phenological stage is relevant. Potential cloud cover hinders the acquisition of optical data. Synthetic aperture radar (SAR) data are less affected by cloud cover and are sensitive to crop information like biomass and leaf area index similar to optical vegetation indices (Mattia et al., 2003; Della Vecchia et al., 2008; Inoue et al., 2014). Consequently, SAR sensors could be a reliable data source for monitoring phenology and for detecting different phenological stages (Lopez-Sanchez et al., 2012; Vicente-Guijalba et al., 2015; Mascolo et al., 2016; Wang et al., 2019). Phenological stages of rice were detected in a threshold classification approach with different polarimetric features at X-band (Lopez-Sanchez et al., 2012). Polarimetric features from Radarsat-2 at C-band were used in order to detect the phenological stages of barley, wheat and oat with Kalman filters (Vicente-Guijalba et al., 2015) as well as with different classification frameworks such as complex Wishart classifier (Mascolo

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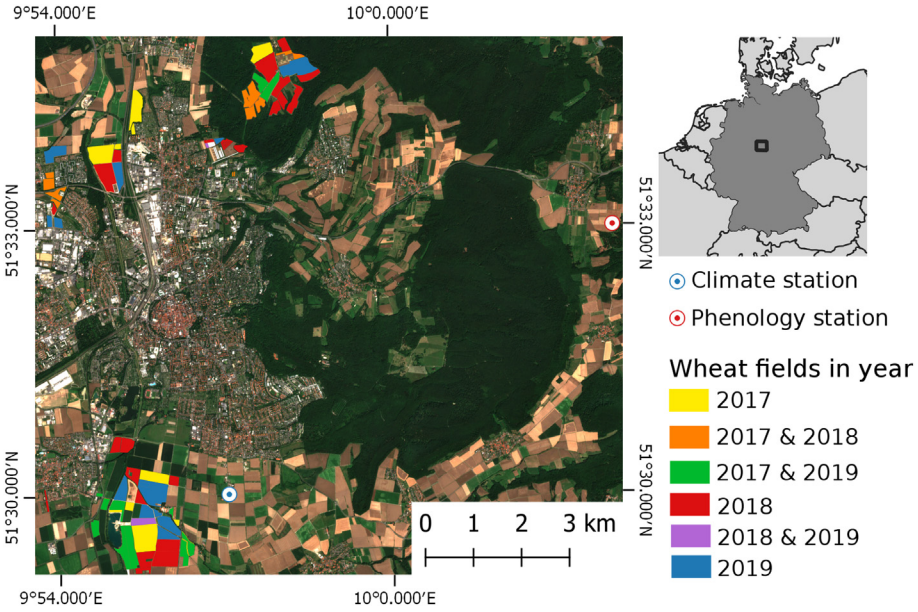


Fig. 1. Overview of study area Göttingen with true color composite of Sentinel-2 from July 24, 2019.

et al., 2016) and machine learning (Wang et al., 2019). These studies attempt to detect the correct phenological stage at a certain point in time and achieved high accuracies (Lopez-Sanchez et al., 2012; Vicente-Guijalba et al., 2015; Mascolo et al., 2016; Wang et al., 2019). It is worth noting that none of them utilized time series metrics in order to detect the date of the phenological stage, which are commonly used in optical remote sensing (Forzieri et al., 2010; Pan et al., 2015; Guo et al., 2016).

Studies in the past demonstrated that C-band backscatter in particular is sensitive to the phenological cycle of different crops based on their sensitivity to crop parameters like aboveground biomass (Skriver et al., 1999; Mattia et al., 2003; Balenzano et al., 2011; Veloso et al., 2017). In addition, the Sentinel-1 sensors potentially acquire dense time series data every 6 days with the same acquisition properties based on their independence on cloud coverage as well as their sensor constellation (Torres et al., 2012). Consequently, they have a high potential for extracting the dates of phenological stages (Mascolo et al., 2016; Veloso et al., 2017), whereas time series analysis of SAR data analysis have remained to date for this purpose underexplored.

The objective of this study was to analyze dense Sentinel-1 time series data in order to extract the date of shooting, ripeness and harvest. The detected dates of these phenological stages from Sentinel-1 time series metrics were compared with in situ observed dates for each phenological stage. This was assessed in selected winter wheat fields in Göttingen, Germany, because wheat is the most important cereal crop in Europe and the cereal crop with the second largest coverage worldwide (Trnka et al., 2014). In addition to the different polarimetric backscatter coefficients, a polarimetric ratio and the interferometric coherence were assessed, which were also considered sensitive to crop parameters (Engdahl et al., 2001; Brown et al., 2003; Della Vecchia et al., 2008; Veloso et al., 2017). The extraction of temporal metrics from the dense Sentinel-1 time series data were applied in three different years, which had substantial differences in precipitation. Conditions in the study area in 2017 were relatively moist, whereas 2018 and 2019 were dry. The meteorological conditions impact the start and end date of phenological stages. It can be expected that the behavior of the backscatter coefficient was different over the analyzed years. Therefore, another objective of this study was to assess if the different phenologies caused by different meteorological conditions are detectable with the Sentinel-1 time series data.

2. Study site & data

2.1. Sentinel-1 data

The Sentinel-1 data were used to create time series for three different years; 2017, 2018 and 2019. The Sentinel-1 mission provides data with a repeat cycle of 12 days using one satellite and 6 days using the two Sentinel-1 satellites (Torres et al., 2012). The analysis focused on the vegetative period from March to the end of August. A total of 93 (31 per year) single look complex (SLC) datasets were available for the three consecutive study periods in the study area. Note that only the same orbits with the relative orbit number 66 were used in order to enable interferometric capabilities.

The Interferometric Wide Swath mode with VH (vertical-horizontal) and VV (vertical-vertical) polarization is the main operational mode of the Sentinel-1 mission (Torres et al., 2012), which was also used in this study. The incidence angle ranged from about 30 to 46 . The resolution of all scenes was about 3 m in range and 22 m in azimuth and had a pixel spacing of 2 m in range and 14 m in azimuth.

2.2. Auxiliary data

Agricultural field information was provided by the Reinshof Experimental Farm of the University of Göttingen. This data included the field boundaries and crop type in the respective years. Fields with wheat that were larger than one hectare were used in this study. In total, 94 wheat fields were analyzed, where wheat was cultivated on 33 fields in 2017, 35 in 2018 and 26 in 2019 (Fig. 1). The size of the fields ranged from 1.1 ha to 27.1 ha, where the average size of the fields was 6.6 ha with a standard deviation of 5.5 ha.

The observed phenological stages were retrieved from the Climate Data Center of the Deutscher Wetterdienst (DWD). This data were collected at sample points from a dense national monitoring network, which is then quality controlled by the agency (Kaspar et al., 2015). In this study, the closest phenological observation was used, which was on average 10.4 km away from the fields, where the closest field was 7.0 km and the farthest field was 14.0 km away from the station (Fig. 1). A sample of 20% of the wheat fields were visited regularly in 2019 in order to confirm the phenological dates.

Four phenological stages were of interest in this study based on their importance for monitoring of ecosystem services (Conradt et al., 2015;

Möller et al., 2017, 2019). These were the beginning of the shooting, mil ripeness, yellow ripeness and harvest. The beginning of the shooting occurred when the majority of the plants began their stem elongation and the first node was 1 cm above the tillering node. This stage has the *Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie* (BBCH) scale 31 (see also Meier (2018)). The grains reached their final size but were still green with milky content in mil ripeness (BBCH = 75), whereas the grains were hard and solid in the yellow ripeness (BBCH = 87). Their time difference was about 17 days in all years.

Climate data provided by the Climate Data Center of the Deutscher Wetterdienst (DWD) was extracted from the Göttingen station for the years 2017, 2018 and 2019. The climate station is located south of Göttingen close to the southern analyzed fields (Fig. 1). The distances to the weather station from the study fields ranged from 1.2 km to 13.3 km, with an average distance of 6.2 km. Consequently, the extracted temperature and precipitation were considered representative for the fields assuming that the temperature and rainfall pattern did not change substantially at the maximum distance of about 13 km. Mean temperature and sum of precipitation both five days before each Sentinel-1 acquisition and on the acquisition date itself were extracted in order result in a similar time scale compared to the Sentinel-1 revisit time.

2.3. Study site

The study area is located in central Germany near to the city of Göttingen, where the predominant land use in this area is agriculture (Fig. 1). The main crops are winter crops (e.g. winter wheat) and corn. In general, the area is relatively flat. The Leine river runs through the study site resulting in mainly Fluvisol and Phaeozem soils, which are considered very fertile (WRB, 2014).

The climate in the study site is temperate with cool winters and warm summers. Normally, the precipitation is distributed relatively evenly over the year. The annual mean temperature is about 9.5 C and the annual mean precipitation sum is about 622 mm. However, the climate of the years 2017, 2018 and 2019 was substantially different. The total precipitation in 2018 and 2019 (184 mm and 316 mm, respectively) was low compared to 2017 (512 mm) over the March to August period (Fig. 2).

3. Methods

3.1. Sentinel-1 processing

The Sentinel-1 TOPSAR (Terrain Observation by Progressive Scan) data consisted of multiple sub-swaths, which included a series of bursts (Torres et al., 2012). The study focussed on one sub-swath, which was split from the other sub-swaths and its individual bursts were merged (i.e. debursted) for the retrieval of backscatter coefficient γ^0. The Sentinel-1 data were multi-looked with 6 looks in range and 2 looks in azimuth to achieve an approximately squared pixel spacing of about 20 m and to reduce speckle noise. The Sentinel-1 data were calibrated to β^0 based on the area normalization factor ${A_{dn}}^2$ and the calibration constant K (ESA, 2015)

$$\beta^0 = \frac{DN^2}{A_{dn}^2 K} \tag{1}$$

Furthermore radiometric terrain flattening was applied in order to obtain γ^0 and to minimize topographic effects (Small, 2011)

$$\gamma^0 = \beta^0 \frac{A_\beta}{\int_{\mathrm{DEM}} A_\gamma} \tag{2}$$

where A is the respective reference area and DEM is a digital elevation model, where the Shuttle Radar Topography Mission (SRTM) DEM at 1 arc-second resolution was used consistently for all images. The single images acquired in the same year were co-registered and a multi-channel filter was applied in order to decrease speckle noise (Quegan and Yu, 2001; Schlund et al., 2018)

$$J_k\left(x, y\right) = \frac{\hat{\sigma}_k(x, y)}{M} \sum_{i=1}^{M} \frac{I_i(x, y)}{\hat{\sigma}_i(x, y)}, 1 \le k \le M \tag{3}$$

The result of this filter was a multi-channel filtered image J_k on position x and y for each input image I_i, where M is the number of images. The local mean intensity $\hat{\sigma}_i(x, y)$ was estimated with the adaptive Lee Sigma filter and a pixel kernel size of 5 by 5 in order to preserve edges. This filter was applied in multi-temporal domain using all acquisitions from the respective vegetation period. Furthermore, the filter was applied in the dual-polarized domain assuming that speckle in the VH and VV are uncorrelated thus reducing the speckle noise further (Quegan and Yu, 2001; Schlund et al., 2017; Schlund et al., 2018). The ratio of VH and VV polarization was also calculated (note that it was calculated in decibel [dB] scale and thus, is calculated as the difference)

$$\gamma^0_{\mathrm{Ratio}} = \gamma^0_{\mathrm{VH}}[\mathrm{dB}] - \gamma^0_{\mathrm{VV}}[\mathrm{dB}] \tag{4}$$

which was considered suitable for crop monitoring in the past (Della Vecchia et al., 2008; Veloso et al., 2017; Homayouni et al., 2019). All images were geocoded with the same DEM and resampled to 10 m pixel spacing.

In the interferometric coherence processing, the Sentinel-1 TOPSAR data were split similarly to the backscatter processing. The single images were co-registered, where the acqusition from one date was used as master and the acquisition 6 days later was used as slave. Consequently, all interferometric products were based on a 6-day temporal baseline. It was assumed that longer baselines would generally result in lower coherences and thus, would not include more information compared to the 6-day temporal baseline. Interferograms of VH and VV polarization respectively were calculated and again a multi-looking with 6 looks in range and 2 looks in azimuth was applied. The interferometric coherences were estimated from this interferograms and debursted. Finally, all coherences in VH and VV polarization were geocoded with the SRTM 1 arc-second DEM and resampled to 10 m pixel spacing similar to the backscatter coefficient. The Sentinel-1 data were processed with the Sentinel Application Platform (SNAP).

3.2. Extracting metrics from Sentinel-1 time series data

The mean value of the backscatter coefficient γ^0, the ratio as well as the interferometric coherence in VH and VV polarization in each acquisition date was extracted for every individual winter wheat field. In addition, mean values of acquisition dates for all wheat fields were also calculated in order to compare them over the three years. The data were smoothed in the temporal domain with a local polynomial regression fitting, namely LOESS (locally estimated scatterplot smoothing) (Cleveland et al., 1992). Such local polynomial regression fittings are commonly used in remote sensing time series data analysis (Verbesselt et al., 2010, 2012). A one degree polynomial regression (i.e. locally linear) was fitted with the weighted least squares method for each backscatter and coherence value and its neighbors, where the neighbors were weighted according to their distance to the value (Cleveland et al., 1990, 1992). A tri-cubic weight function w was used for the weighting of the neighbors

$$w = \left(1 - \left(\frac{d}{\max(d)}\right)^3\right)^3 \tag{5}$$

where d is the distance of a given value to the fitted value (Cleveland et al., 1990). The size of the neighborhood was defined by the span, which had a substantial influence on the smoothness of the fitting.

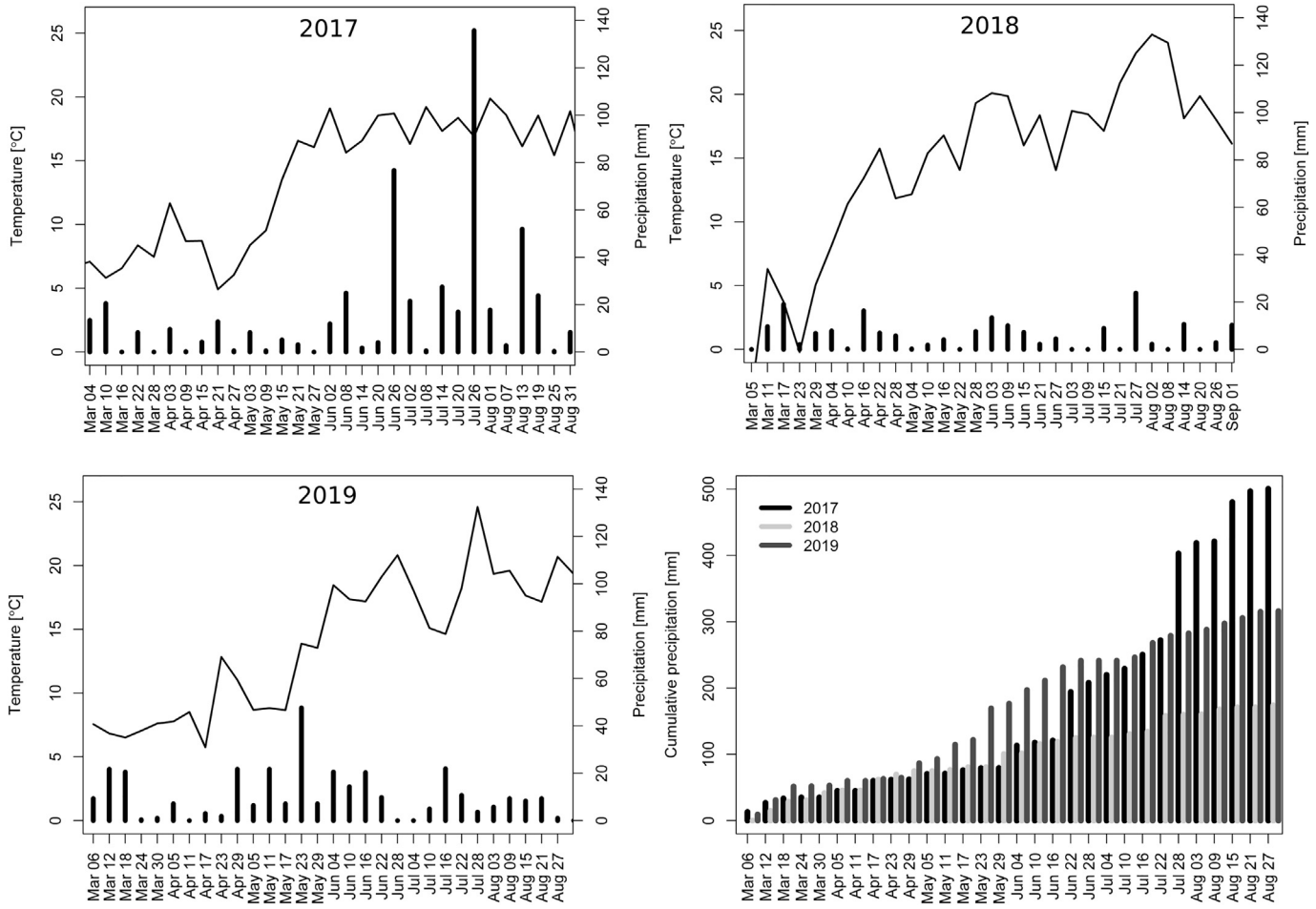


Fig. 2. Temperature (°C) and precipitation (mm) during Sentinel-1 acquisitions for the years 2017 (upper left), 2018 (upper right) and 2019 (lower left) as well as a comparison of cumulative precipitation for the years 2017, 2018 and 2019 (lower right).

Therefore, the fitted curves were visually inspected and a generalized cross validation was used in the fit of the mean values of the each acquisition dates in all wheat fields (Golub et al., 1979). Finally, the span value which minimized the generalized cross validation error was used for all wheat fields as well as for the individual wheat fields. The span value was 0.4 for the backscatter coefficients and 0.3 for the coherences in both polarizations in the three years. The local polynomial regression fitting ensured that single-date phenomena had only a small effect on the time series.

The smoothed data were used to extract different metrics. First, local minima and maxima were calculated for the fitted curves in each available wheat field. The point x was defined as a local maxima as $f(x) \geq f(x_0)$, where x_0 was the point next to x. Similarly, the point x was defined as a local minima as $f(x) \leq f(x_0)$.

In addition to the local extrema, breakpoints were derived where the regression (i.e. the time series) was split iteratively into different segments. The breakpoints were detected where the regression coefficients differ from one segment to the next one. The optimal position of the breakpoints was retrieved by minimizing the residual sum of squares (RSS) of the respective segments (Bai and Perron, 2003; Zeileis et al., 2003). The detection of breakpoints was frequently used in time series analysis studies, for example to detect deforestation (Verbesselt et al., 2010, 2012) and trend changes in forests (Forkel et al., 2013). It is assumed that (substantial) changes, like the transition/growing of plants or the harvesting of the wheat fields, result in breakpoints in time series and thus are detectable with such an approach.

The temporal metrics (i.e. local extrema and breakpoints) were computed for the VH and VV backscatter coefficient and their ratio (VH/VV) as well as for the VH and VV interferometric coherence in all wheat fields. The dates when the temporal metric was detected were also retrieved. The detected dates with Sentinel-1 time series metrics t_{Det} were compared to the observed phenological stage dates t_{Obs}

$$t_{\mathrm{Diff}} = t_{\mathrm{Obs}} - t_{\mathrm{Det}} \tag{6}$$

The time series analysis was implemented in the R environment (R Core Team, 2019).

3.3. Reduction of temporal resolution

Sentinel-1 A and B data were used in this study to achieve the temporal resolution of 6 days. However, the 6-day temporal resolution is mainly ensured over Europe and many other parts in the world are covered with a 12-day revisit cycle. Only Sentinel-1 A data were used to create an additional time series with a 12-days temporal resolution. This simulated a time series without a second satellite in space or the temporal resolution in most parts of the world. Similar to the full time series data, the lower resolved time series was smoothed with a local polynomial regression fitting and the temporal metrics were retrieved. The same metrics were used to indicate phenological stages and were compared to the in situ observed dates of the phenological stages. This analysis indicated the impact of temporal resolution on the accuracy of the phenological stage indication with Sentinel-1 time series data.

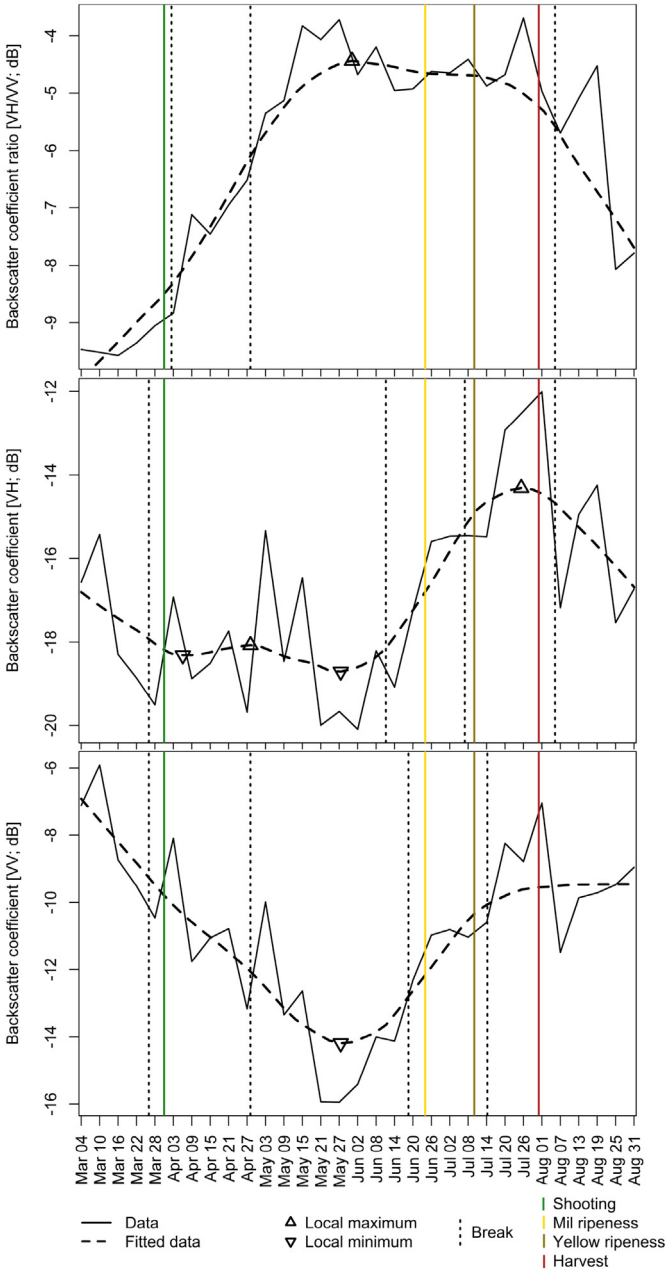


Fig. 3. Backscatter coefficient ratio VH/VV (top), backscatter coefficient in VH (center) and VV (bottom) polarization over time for one wheat field in 2017 with smoothed data and detection of local maxima, local minima and breakpoints.

4. Results

4.1. Temporal accuracy of Sentinel-1 backscatter time series metrics

4.1.1. Accuracy of backscatter time series metrics with 6-day temporal resolution

The results for the time series of backscatter coefficients and their ratio are shown for one sample wheat field for 2017 (Fig. 3). The backscatter coefficient generally decreased from March to June in both polarizations, where the decrease was higher in VV compared to VH polarization. A local minimum was detected at the end of May in both polarizations. The backscatter coefficient increased to a local maximum during July when the full ripeness of the wheat plants was reached. The harvest at end of July resulted in a decrease in VH polarization again, whereas the VV polarization remained similar. The different degree of decrease and increase of the polarizations resulted in a backscatter coefficient ratio which increased substantially with the beginning of the shooting and remained stable over summer until the senescence and finally harvest. An increase of the ratio means less difference between VH and VV polarized backscatter coefficients. The ratio decreased (i.e. the difference between VH and VV increased) again at the harvest, resulting in a breakpoint detection (Fig. 3).

The breakpoints of backscatter coefficient ratio were used to detect the shooting and the harvest. The time difference between the first breakpoint of VH/VV ratio in the vegetation period and the date of the shooting was about 4.6 days in 2017, 5.3 days in 2018 and 9.5 days in 2019. The detection of the shooting was generally earlier compared to the true date in 2017 and 2019, whereas it was the opposite in 2018 (Fig. 4a). The spatial representation of the temporal accuracy in the detection of the shooting also revealed that there was no spatial pattern and the majority of the absolute temporal differences were less than 10 days (Fig. 5).

The root mean squared difference between the last breakpoint of the ratio and the harvest was 5.1 days in 2017, 8.2 days in 2018 and 10.4 days in 2019 (Fig. 4b). It is worth noting that the ratio resulted in the same temporal difference in all fields at 5 days (except two fields with a temporal difference of 2 days) in 2017. In contrast, the range of temporal differences was generally larger in 2018 and 2019. Nevertheless, the median of the temporal differences between the breakpoint indicating the harvest and the date of the harvest was 3 days in 2018 and 0 days in 2019 (Fig. 4b).

The local maximum of VH backscatter after the shooting was generally at the time of ripeness of the wheat fields. However, the local maximum in 2017 was generally later compared to 2018 and 2019. The temporal difference between the local maximum of VH backscatter coefficient and mil ripeness was 5.7 days, whereas the median was about 0 (Fig. 6). However, the same metric displayed large differences to the date of mil ripeness of about 34.4 days in 2017 and 21.5 days in 2019. In general, the difference was negative indicating that the local maximum was later than the mil ripeness. It indicated the time of yellow ripeness or even harvest in 2017 (Fig. 3). The temporal difference between local maximum and yellow ripeness decreased by a few days for the years 2017 and 2019, whereas it increased substantially in 2018 compared to the mil ripeness (Fig. 6).

4.1.2. Accuracy of backscatter time series metrics with 12-day temporal resolution

As expected, the lower temporal resolution resulted in lower temporal accuracy in the detection of the phenological stages with Sentinel-1 backscatter time series metrics. The accuracy when using the first breakpoint in the vegetation period as an indication of the shooting of the plants was 5.5 days in 2017, 9.1 days in 2018 and 9.6 days in 2019 (Fig. 7). The decrease in accuracy was less than one day in the years 2017 and 2019, whereas it was about 4 days in 2018. A similar pattern was observed in the indication of the harvest. The difference of the temporal accuracy was again one day comparing the different temporal resolutions for the years 2017 and 2019, where more outliers (indicated by the points in the boxplots) were observed compared to the 6-day temporal resolution in 2018 (Figs. 4 & 7). Similar to the 6-day temporal resolution, the range of temporal differences was larger in 2018 and 2019 compared to 2017. Nevertheless, the median in 2018 was again 3 days and the difference between the 6-day and 12-day temporal resolution was marginal (0.4 days).

4.2. Temporal accuracy of Sentinel-1 coherence time series metrics

The coherence was about 0.7 in VH polarization and 0.8 in VV polarization at the beginning of March in the sample field, where the plants were relatively small. The coherence decreased from March to June to 0.3 in both polarizations. It remained low over the summer and increased again in August after the harvest. In general, the coherence of

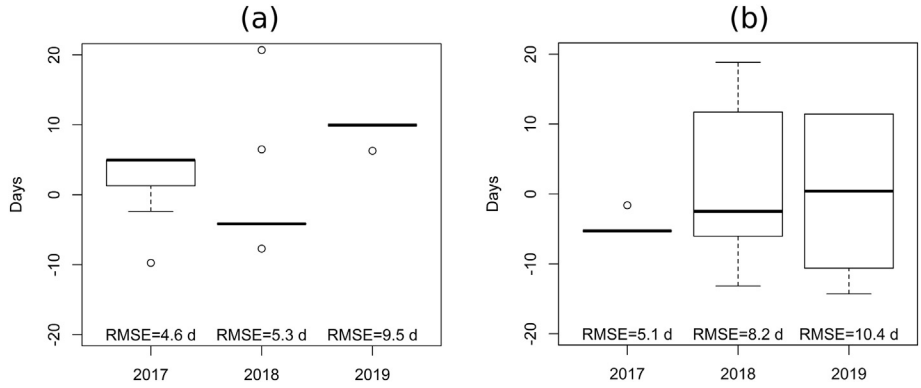


Fig. 4. Boxplots of the temporal differences between the first detected breakpoint in VH/VV backscatter coefficient ratio and the shooting date (a; left) as well as the temporal differences between the last detected breakpoint in VH/VV backscatter coefficient ratio and harvest (b; right) in the years 2017, 2018 and 2019.

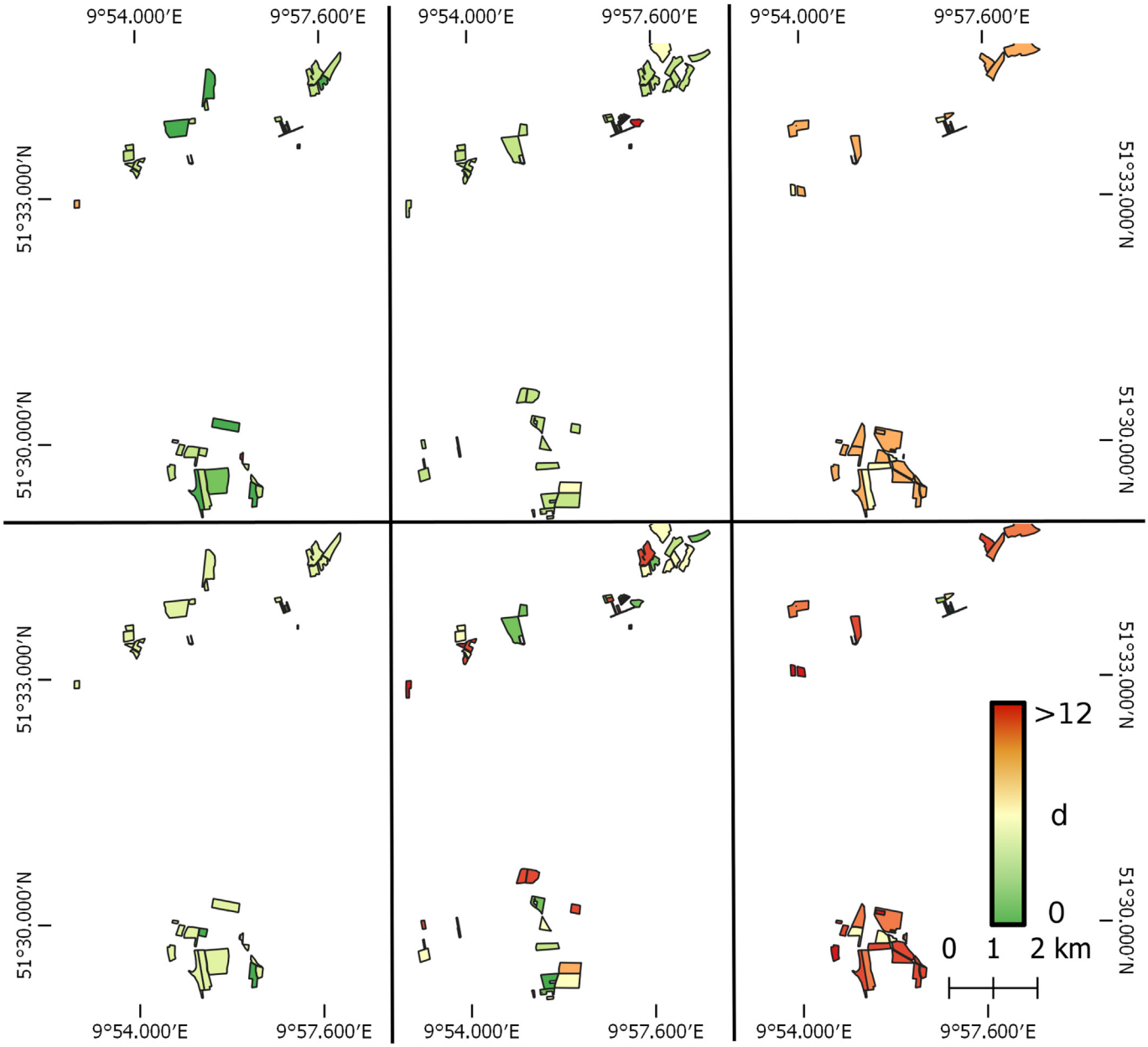


Fig. 5. Spatial representation of the absolute temporal differences between the first detected breakpoint in the VH/VV backscatter coefficient ratio and the shooting date (top) as well as the temporal difference between the last detected breakpoint in the VH/VV backscatter coefficient ratio and harvest (bottom) in the years 2017 (left), 2018 (center) and 2019 (right).

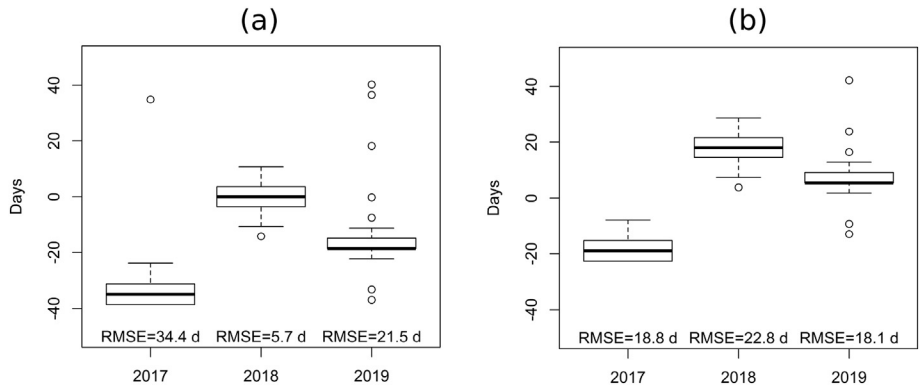


Fig. 6. Boxplots of the temporal differences between the local maximum after shooting in VH backscatter coefficient and date of mil ripeness (a; left) as well as the temporal differences between the local maximum after shooting in VH backscatter coefficient and date of yellow ripeness (b; right) in the years 2017, 2018 and 2019.

VH and VV polarizations had a similar pattern (Fig. 8).

The interferometric coherence was only able to indicate the date of the shooting of plants. This was indicated by the first breakpoint in the vegetation period. That was similar to the VH/VV backscatter coefficient ratio. The temporal accuracy was about one week in all three analyzed years using the VV polarization. The majority of the fields had the same detected dates independent of the polarization, but VH polarizations had more outliers compared to VV polarization. Therefore, the temporal accuracy was substantially lower in VH compared to VV polarization (Fig. 9).

4.3. Comparison of the three years

The results suggest that the backscatter and especially the backscatter ratio between VH and VV were able to detect the beginning of the vegetation period with the shooting and ending with the harvesting of the plants. Therefore, the three years were compared, which were substantially different in their precipitation pattern (Fig. 2). The general pattern of the different polarized backscatter coefficients and their ratio was generally similar over the three years (Fig. 10). It is important to note that the range of backscatter coefficients were different in the three different years. In contrast, the backscatter coefficient ratio had similar values over the three years. Nevertheless, the decrease of ratio occurred substantially earlier in the years 2018 and 2019 compared to 2017. This was based on the fact that the maximum and decrease of VH backscatter coefficient was substantially earlier in those years compared to 2017 (Fig. 10).

5. Discussion

The ratio of VH and VV polarized backscatter had a large potential for estimating the phenological stages of winter wheat. This was observed for the shooting and harvest period. In the past, it was suggested that VH and VV polarized backscatter ratios are related to crop parameters like biomass and thus have the potential for monitoring crops (Della Vecchia et al., 2008; Homayouni et al., 2019). The backscatter was relatively similar for both polarizations until the end of March before the shooting begins. It is assumed that the backscatter is dominated in both polarizations by soil properties like roughness and moisture, where the shooting and elongation of the plants in the vertical dimension resulted in an attenuated backscatter (Skriver et al., 1999; Veloso et al., 2017). The attenuation of the signal, due to the vertical growth of stems, was substantially different between the polarizations (Balenzano et al., 2011), which caused a change and the break in the VH/VV ratio. This resulted in the detection of shooting with the first breakpoint in the backscatter ratio time series. It is worth noting that most winter wheat fields displayed the same date of the shooting thus confirming the stability of the cross-polarized ratio over various fields and conditions.

In general, ratios or differences can be considered generally more insensitive to moisture and incidence angle variations compared to individual polarizations because such effects are to a certain extent similar for both polarizations and thus would be reduced in the difference or ratio (Brown et al., 2003; Della Vecchia et al., 2008; Schlund and Davidson, 2018). This was confirmed by the similarity of the backscatter coefficient ratio over the three years despite their substantial difference in precipitation.

The volume scattering increased in June when the plants develop to ripe stages (Skriver et al., 1999). The backscatter increased in both polarizations at the same time due to the increase of plant biomass (Mattia et al., 2003; Veloso et al., 2017). A decrease of backscatter can be expected in the ripeness stage due to the lower water content in the plants after the mil ripeness (Veloso et al., 2017). It can be assumed that the different weather conditions between the three years caused variability in the backscatter coefficients. Therefore, it was not possible to detect the same phenological stage with the local maximum

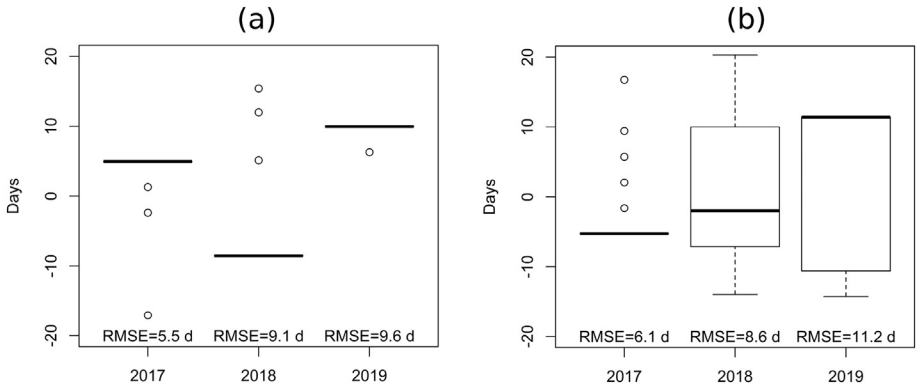


Fig. 7. Boxplots of temporal differences between the first detected breakpoint in the VH/VV backscatter coefficient ratio of the 12-day Sentinel-1 time series and shooting date (a; left) as well as the temporal differences between the last detected breakpoint in the VH/VV backscatter coefficient ratio of the 12-day Sentinel-1 time series and harvest (b; right) in the years 2017, 2018 and 2019.

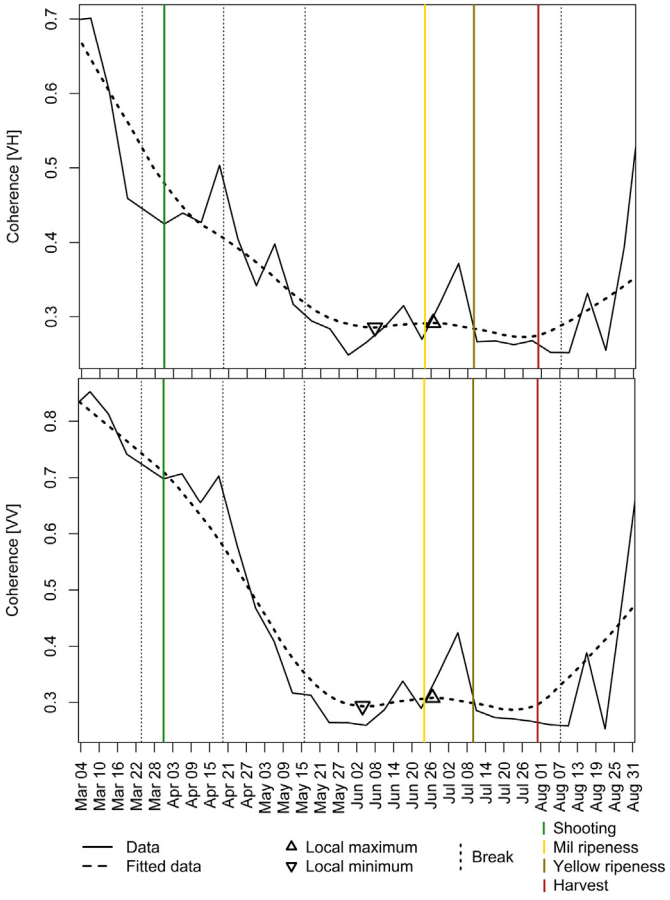


Fig. 8. Interferometric coherence in VH (top) and VV (bottom) polarization over time for one wheat field in 2017 with smoothed data and detection of local maxima, local minima and breakpoints.

throughout the three studied years. The VH/VV ratio had similar values over the ripeness, where only a small decrease was observed.

It was expected that the backscatter decreased substantially after the harvest, but this was not true for all fields. This was based on the fact that stubble potentially results in a similar backscatter compared to ripe crops depending on the residual cover (Bouman and van Kasteren, 1990; McNairn et al., 2002). In addition, the post-harvest practices differed between fields and thus the detection of harvest is a difficult task and requires more attention in future studies (Bouman and van Kasteren, 1990). Again, the backscatter ratio had a greater potential for detecting the harvest with a breakpoint in the time series compared to the individual polarizations. This was based on the fact that the attenuation and dominant backscatter contributions were substantially different in the two polarizations. No or low volume contribution existed after the harvest and thus the VH backscatter decreased, whereas the direct contribution from the soil dominated again in VV polarization resulting in a similar backscatter compared to pre-harvest acquisitions.

In general, the detection of dates of phenological stages resulted in a level of accuracy calculated as RMSE of about 6 days or less. This is better than the temporal resolution of Sentinel-1. As expected, the decrease of temporal resolution to 12 days resulted in a less accurate estimation of the dates. However, the level of accuracy was still below the temporal resolution of the used time series (i.e. 12 days). These results suggest that increasing the temporal resolution will further improve the level of accuracy. The combination of different orbits is necessary to achieve a higher resolution with Sentinel-1, where the effect of different incidence angles and viewing directions should be studied further. It can be argued that the VH/VV ratio is less sensitive to varying conditions like different incidence angles and thus it is useful in the combination of different viewing geometries in order to obtain a higher temporal resolution (Brown et al., 2003; Della Vecchia et al., 2008).

The calculation of the interferometric coherence is not possible with different viewing geometries. However, the interferometric coherence was only able to detect the shooting, where the temporal accuracy was generally lower than the detection with VH/VV backscatter ratio. It could be expected that the coherence increases substantially after the harvest based on the absence of volume decorrelation. Nevertheless, the coherence remained low in many fields, which hindered the detection of harvest with interferometric coherence independent on the polarization similar to the individual backscatter coefficients. This was based on the fact that different conditions during the acquisitions as well as different post-harvest practices and stubble could result in low coherences.

The backscatter coefficients and their ratio as well as the coherences showed generally a similar behavior over the three different years. However, the timing of phenological stages was substantially different, where the plants reached senescence earlier in 2018 and 2019. This was also indicated especially by the VH/VV ratio, which decreased substantially earlier in 2018 and 2019 compared to 2017. It is worth noting that they were on equal levels in the three years confirming again the stability of the VH/VV ratio. In contrast, the backscatter coefficients were substantially different in the three years due to the different weather conditions, which also resulted in different vegetation development. Different studies suggested that the backscatter is correlated with the biomass and/or leaf area index of different crops (Mattia et al., 2003; Della Vecchia et al., 2008; Inoue et al., 2014; Homayouni et al., 2019). Further studies should investigate if the different backscatter responses indicate a different biomass and consequently yield in these three different years. The times series metrics presented in this study

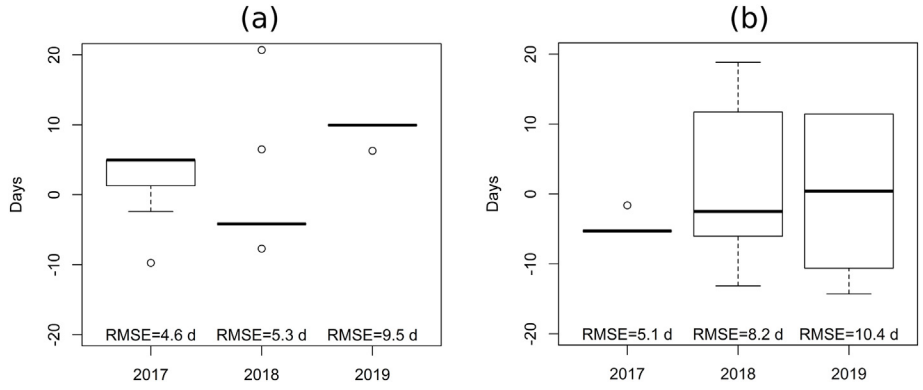


Fig. 9. Boxplots of the temporal differences between the first detected breakpoint in VH coherence and the shooting date (a; left) as well as the temporal differences between the first detected breakpoint in VV coherence and shooting (b; right) in the years 2017, 2018 and 2019.

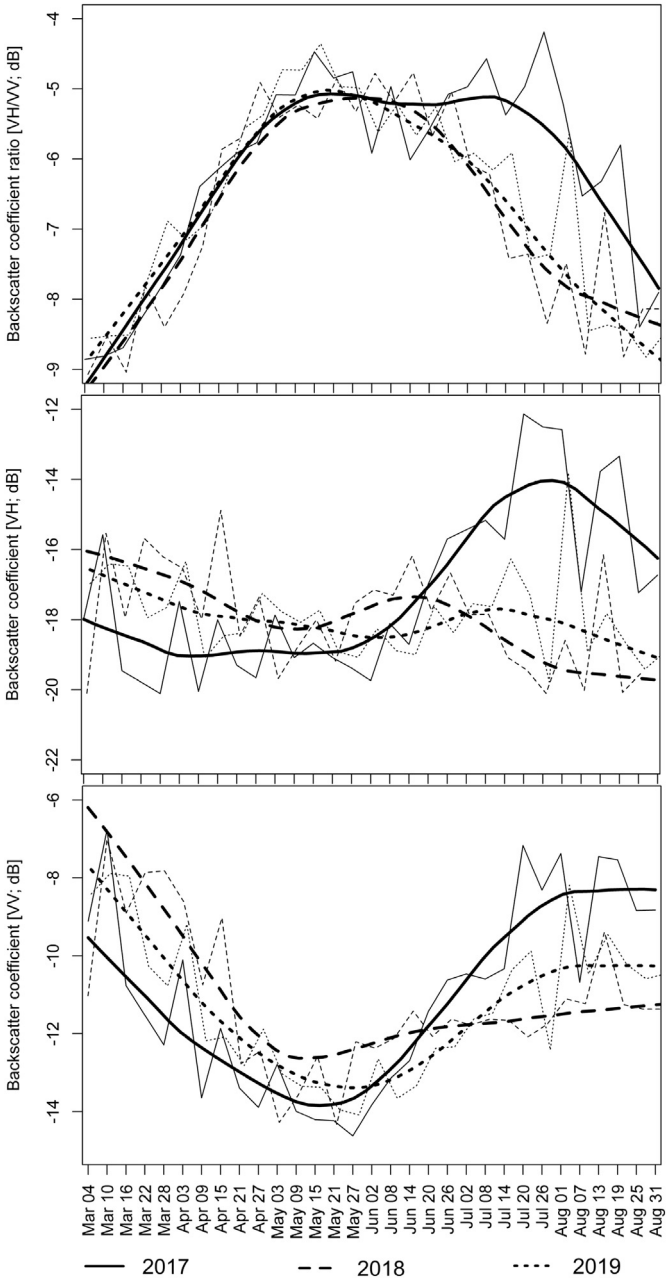


Fig. 10. Backscatter coefficient ratio VH/VV (top), backscatter coefficient in VH (center) and VV (bottom) polarization over time for all wheat fields in 2017, 2018 and 2019.

indicate at minimum the vegetation period, which was substantially different across the three years.

Optical data were also used in the estimation of phenological stages in the past (Arvor et al., 2011; Pan et al., 2015; Guo et al., 2016; Veloso et al., 2017). It can be assumed that the moist year 2017 also had long periods of cloud cover. This hinders the acquisition of optical data and the estimation of phenological stages. SAR data can be used to fill the gaps in cloud covered optical time series data, where both data sources are sensitive to the phenology of agricultural crops (Veloso et al., 2017; d'Andrimont et al., 2020). In addition, it can be assumed that the sensitivity of optical data to biomass and thus potentially to phenology decreases earlier in strong covered crops due to the saturation of the signal than does the SAR data (Veloso et al., 2017). Consequently, Sentinel-1 can be used alone or in combination with optical data in order to estimate spatially and temporally consistent phenological information of winter wheat.

6. Conclusion

Dense time series and their derived metrics are frequently used to detect phenological stages with optical remote sensing (Arvor et al., 2011; Pan et al., 2015; Guo et al., 2016; Veloso et al., 2017). This study confirms that SAR sensors could be used to detect phenological stages of crops based on time series metrics, which could be used to monitor crops on a regular basis. The weather and sun independence of such sensors could be an important advantage in order to provide a dense time series. For instance, the Sentinel-1 mission ensures a 6-day temporal resolution over Europe using the repeat-pass orbit. The results suggest that the backscatter coefficient ratio has a high potential for detecting the shooting and the harvesting of wheat fields. The value range of the ratio was relatively stable over time, which indicates its insensitivity to different environmental conditions. Therefore, Sentinel-1 time series and especially its backscatter coefficient ratio is useful in a crop monitoring system and could provide information in order to develop adaption strategies to weather phenomena and changing environmental conditions.

CRediT authorship contribution statement

Michael Schlund:Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft.**Stefan Erasmi**:Conceptualization, Methodology, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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